

How much edit experience should the model receive?

Sequence-edit JEPA validity audit and bounded pilot

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The task is synthetic typo repair

- ▶ Official iGSM supplies a problem and its clean, multi-step worked solution.
- ▶ We damage that solution with token insertions, deletions, and replacements.
- ▶ The expert path is the exact reverse of those corruptions, so it is oracle denoising rather than naturally observed editing.
- ▶ More independent damaged solutions add breadth; more alternative edits per state add local action coverage.

Why it matters: we must establish healthy primitive dynamics before asking whether hierarchy helps.

The old hierarchy round is invalid

- ▶ All five jobs failed before their first optimizer step, so they provide no architectural evidence.
- ▶ The immediate symptom was a 276-token “sentence” exceeding a 128-position table.
- ▶ The cause was deeper: official steps often end in fused tokens such as 6., while the adapter split only on standalone ..
- ▶ The corrected adapter preserves official nested steps and exactly reverses every audited corruption.

What to notice: this is an invalid result, not evidence against hierarchy.

The corrected data contract passes its first gate

Audit check	Observed result
Exact terminal recovery	128/128
Multi-step solution collapse	0/128
Official solution length	median 166.5 tokens
Official steps	median 9
Maximum step length	p95 43 tokens
Repairs per trajectory	mean 11.14
Repair/minimum-distance ratio	mean 1.029

Why it matters: the model now sees the intended step sequence, but these are data-validity observations, not trained-model results.

The next pilot varies one data axis at a time

- ▶ The $K=0$ data curve uses 512, 2,000, and 6,000 unique trajectories for three passes.
- ▶ The counterfactual curve uses $K=0, 1, 4,$ and 8 exact alternative outcomes at the common 2,000-trajectory anchor.
- ▶ Alternatives use observed current-buffer tokens and carry no preference, defect, or remaining-distance label.
- ▶ A 128-example H4 cell is only a full-shape process smoke; architecture comparisons remain gated.
- ▶ We retain the smallest K near the best result only if it improves held-out dynamics without factual-error or rank regression.

Next decision: spend later compute on unique diversity, local alternatives, or neither.